

# Adapting the Actor Model of Concurrency for High-Frequency Trading

Synchronous Message Delivery (`fast_send`) and a Tick-to-Book Latency Study

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## Abstract

The many benefits of the actor model — per-actor state isolation, freedom from data races, and sequential single-message reasoning — have been recognized for decades, yet the model has long been dismissed as unsuitable for high-frequency trading (HFT). The reasoning was that actors imply many threads, many mailbox queues, and a message — with its heap allocation, queue operations, scheduler dispatch, and context switch — for every interaction: overhead incompatible with a microsecond latency budget. This paper argues that the dismissal is wrong for co-located actors, and supports the argument not only analytically but with a deployed, measured implementation — *kaspar-hft*, an open-source C++20 framework that adapts the actor model into a form suitable for HFT and is its reference implementation. The extensions are four: **fast\_send**, a synchronous delivery mechanism in which the sending thread runs the receiver’s handler inline and takes the reply as a value; actor *groups*, which co-schedule a set of actors on one thread behind a single shared mailbox; selectable mailbox queue implementations matched per actor to its write pattern; and a memory pool that removes per-message allocation overhead.

The central mechanism is **fast\_send**, whose defining property is *receiver transparency*: the handler is written identically for synchronous and asynchronous delivery and does not depend on which was used, which thread executes it, or whether the sender is blocked. Synchronous delivery across a chain of grouped actors runs entirely on the first actor’s thread, reducing scheduler context switches from  $O(N)$  on the asynchronous path to  $O(1)$ ; a thread-local call-chain test detects cyclic synchronous invocation on one thread before any lock is taken, while a cycle spread across threads is outside what it can see and deadlocks; and five delivery operations expose the latency/robustness spectrum behind one transparent interface. Microbenchmarks put the synchronous round trip at tens of nanoseconds — roughly two orders of magnitude below the ungrouped asynchronous path — and a single **fast\_send** dispatch at about the cost of one unpredictable virtual call (about  $5\times$  a direct call).

We then measure the framework in a real HFT setting: socket-to-book (“tick-to-book”) latency on a live CME market-data feed for ES, NQ, and ZN futures, measured with software timestamps from socket read to book update. When the socket-reader thread decodes each packet and updates the book itself, book-update medians are 0.8–1.1  $\mu\text{s}$ , and a  $\sim 9$  ns **fast\_send** hop is about 1% of that. When the reader instead hands each datagram to a second thread through an asynchronous mailbox, the median rises to  $\sim 7$   $\mu\text{s}$ , and the tail grows with in-packet position and, on rare bursts, mailbox-queue backlog, driven by the market’s non-Poisson, strongly clustered arrival process, which we characterize in a companion paper. Finally, the shared-queue group yields a production/simulation duality: the same actor code runs unchanged in live trading and in deterministic backtest. A saturated-throughput and contention evaluation is left to future work.

**Keywords:** actor model, message passing, concurrency, synchronous delivery, deadlock, receiver transparency, low-latency systems.

## 1 Introduction

In the actor model [1, 2], a program is a collection of isolated actors, each with private state, a message handler, and a mailbox. Actors communicate only by asynchronous message passing: a sender places a message on the receiver’s mailbox and proceeds without waiting, and the receiver dequeues and processes messages one at a time on its own thread. This discipline is what gives the model its guarantees — isolation, sequential per-actor processing, and location transparency — and it is essential when actors are distributed across machines.

When the communicating actors reside in the same process on the same machine, however, the asynchronous path carries overhead that the isolation guarantee does not require. A single request-and-reply between two co-located, in-process actors incurs, on conventional implementations:

- **Heap allocation.** Because the message must outlive the sender’s current continuation and persist until processed at an indeterminate later time, it must be heap-allocated, with the associated allocation cost, fragmentation, and garbage-collection pressure.
- **Queue operations.** Each message is enqueued by the sender and dequeued by the receiver, each requiring synchronization on the mailbox.
- **Scheduling and context switches.** The receiver must be scheduled onto a thread to run, incurring a context switch; a context switch typically costs on the order of  $10^3$ – $10^4$  processor cycles and pollutes cache and translation lookaside buffer (TLB) state [31, 32, 33].
- **Round-trip latency.** A reply repeats the entire path in reverse, so a request-and-reply crosses the mailbox twice and the scheduler twice.

For a same-process pair, all of this is avoidable in principle: the sender’s own thread could run the receiver’s handler directly and take the reply as a return value. The difficulty is doing so without breaking the abstraction the actor model exists to provide. A direct handler invocation — calling the receiver’s handler as an ordinary method — discards the receiver’s exclusive-execution guarantee and exposes the receiver as a passive object. The design problem is to obtain the performance of a direct, same-thread call while preserving, from the receiver’s point of view, the semantics of message passing.

The central mechanism this paper introduces is **fast\_send**, a synchronous delivery mechanism that resolves that tension. The sending thread acquires exclusive access to the receiving actor and executes the receiver’s handler directly, on the sender’s thread, returning the reply as a value. The receiver’s handler is identical for synchronous and asynchronous delivery and does not depend on which was used — the property we call *receiver transparency*. Because synchronous invocations can form cycles, and a cycle deadlocks a non-reentrant lock or overflows the stack under a reentrant one, the mechanism carries a lightweight cyclic-invocation detector that fires before any lock is taken.<sup>1</sup> It sees cycles on one thread; a cycle spread across threads needs a different mechanism (Section 7.3). **fast\_send** is implemented in *kaspar-hft*, an open-source C++20 actor framework for high-frequency trading;<sup>2</sup> all code and measurements in this paper are drawn from that repository.

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<sup>1</sup>kaspar-hft does not implement this detector yet: **fast\_send** takes the receiver’s mutex directly, and its only check is a debug-build assertion against an actor synchronously sending to itself, so a synchronous cycle currently deadlocks.

<sup>2</sup><https://github.com/vincent212/kaspar-hft>

**What was changed to make a conventional actor framework HFT-ready.** This is a paper about a working trading system before it is a paper about concurrency: the question it answers is whether a live tick-to-book path can be built on actors without paying a latency penalty for the discipline. Making a textbook actor framework fit that path took four concrete changes, each removing one source of the per-message cost that the mainstream one-thread-per-actor, one-mailbox-per-actor design pays. *First*, **fast\_send**, the synchronous delivery mechanism above, which runs a co-located receiver’s handler inline on the sender’s thread and eliminates the allocation, queue operations, scheduling, and context switch of an asynchronous send. *Second*, actor *groups*, which co-schedule a set of actors on one thread behind a single shared mailbox, so an entire servicing stage collapses onto one core with no cross-thread hop. *Third*, per-actor selection of the mailbox queue implementation, matched to each actor’s write pattern (single bursty producer, high fan-in, or grouped), because no one queue is fastest in every regime. *Fourth*, a memory pool that removes per-message allocation and, more importantly, the multi-millisecond allocator tail that would otherwise land directly in the latency distribution. The remainder of the paper develops each in turn and then measures, on a live CME feed: with decode and book update on one thread, socket-to-book medians are about 1  $\mu$ s, of which a **fast\_send** hop is about one percent.

## Contributions.

1. A synchronous, same-thread actor-delivery mechanism (**fast\_send**) that eliminates the heap allocation, queue operations, scheduling, and context switches of the asynchronous path for co-located actors, and returns the reply as a value (Section 7.1).
2. A precise statement of *receiver transparency*: the handler executes identical code in both modes and does not depend on the delivery mode, the executing thread, or whether the sender is blocked (Section 7.2). It is a property of how handlers are written, not one the framework or the language enforces.
3. A thread-local call-chain membership test that detects cyclic synchronous invocation on one thread before lock acquisition, together with a refined (actor, message-class) variant that admits callback patterns while still catching unbounded recursion (Section 7.3).
4. A family of five delivery operations spanning asynchronous through spin-wait synchronous to best-effort-with-async-fallback, unified behind one transparent handler interface (Section 7.4).
5. Actor *groups*, which co-schedule a set of actors on one thread behind one shared mailbox so that an intra-group send never crosses a core; the single shared queue additionally yields a production/simulation duality, in which the same actor code runs unchanged in live trading and in deterministic backtest (Sections 7.5, 11).
6. Per-actor selection of the mailbox queue implementation over four multi-producer/single-consumer designs, with a selection rule derived from measurement across the grouped, bursty, and high-fan-in regimes (Sections 7.6, 9.2).
7. A memory pool that removes per-message heap allocation and, more importantly, the allocator’s multi-millisecond tail from the latency distribution (Section 7.7).
8. An evaluation of the mechanism in isolation: an analytical account of the per-message work eliminated and microbenchmarks of the C++ implementation, including drain-side batching ( $1.63\times$ ) and memory pooling that together raise asynchronous throughput  $2.46\times$ , and the pool’s two-order-of-magnitude reduction of the latency-tail maximum (Section 9).

9. A live socket-to-book case study on CME market data (Section 10): with decode and book update on the socket-reader thread, book-update medians are 0.8–1.1  $\mu$ s, about 100 $\times$  a `fast_send` hop; with a cross-thread hand-off between reader and decoder, latency decomposes into a floor plus an in-packet serialization slope, and the study separates mailbox-queue occupancy from in-packet position and relates the tail to a non-Poisson, strongly clustered arrival process. A saturated-throughput evaluation is left to future work.

## 2 Origins and history of the actor model

The actor model was introduced by Hewitt, Bishop, and Steiger in 1973 [1] as a single uniform notion — the *actor*, an entity that communicates only by sending messages to other actors — and was given a rigorous semantics over the following decade by Greif [17], Clinger [18], and, most influentially, Agha [2], who recast the model on the primitives *create*, *send*, and *become*. A distinguishing feature of these semantics is *unbounded nondeterminism*: a message is guaranteed to arrive eventually but with no bound on how long it may be delayed, so the set of possible outcomes can be infinite (in contrast to *bounded* nondeterminism, where a fixed process count and prompt scheduling keep that set finite). The most influential early *deployed* system with actor-like structure was Erlang [5], whose designers arrived at similar principles largely independently — Armstrong later placed Erlang closer to communicating sequential processes than to the actor model [19]. The subsequent decades produced a broad ecosystem of actor runtimes — on the JVM, .NET, C++, Rust, Swift, and elsewhere — which we survey next.

## 3 A survey of actor-model implementations

This section surveys the major actor runtimes along the three axes most relevant to `fast_send`: how actors are mapped to operating-system threads, whether and how a synchronous or request–reply path is offered, and what is known of their performance. The consistent findings across the ecosystem — that actors are multiplexed M:N over a small worker pool, and that no mainstream system runs the *receiver’s* handler on the *sender’s* thread — are what locate the contribution of `fast_send`. Table 1 summarizes the comparison; the per-system detail and performance figures follow. The later related-work discussion (Section 5) revisits the systems closest to `fast_send` to sharpen the distinction; here the account is descriptive.

**BEAM: Erlang and Elixir.** The Erlang runtime multiplexes very large numbers of lightweight processes — each with a private heap and mailbox — onto a small set of scheduler threads, one per logical core by default, with a per-scheduler run queue and work-stealing migration [20]. Scheduling is preemptive by a reduction count: a process runs until it consumes its budget (4000 reductions by default) and is then descheduled, which bounds tail latency [20]. The synchronous path, `gen_server:call`, blocks the caller for up to a timeout and returns a value, but the request is placed in the *server* process’s mailbox and handled by `handle_call/3` on the server’s own scheduler; the caller merely performs a selective receive on the reply. The caller’s thread never runs the callee’s handler [5].

**JVM: Akka.** Akka does not bind actors to threads. A dispatcher binds a set of actors to a fork–join thread pool, and an actor occupies a thread only while it has messages to process [7]. A `throughput` setting caps how many messages an actor processes before the thread moves to the next actor; this mailbox batching is the principal latency/throughput lever. Akka’s own

**Table 1.** Actor-to-thread mapping and synchronous messaging across implementations. The final column — whether the sender’s thread ever executes the receiver’s message handler — is negative for every surveyed system and positive only for **fast\_send**.

| System                             | Runtime      | Actor → thread mapping                                                                                         | Synchronous request-reply                                                                       | / Sender runs handler? |
|------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------|
| Erlang/OTP, Elixir                 | BEAM VM      | M:N green processes; one scheduler thread per core; per-scheduler run queues, work stealing                    | <code>gen_server:call</code> blocks caller, returns a value; handler runs in the server process | No                     |
| Akka                               | JVM          | thread-pool dispatchers over a fork-join pool; actors not 1:1 with threads                                     | <code>ask (?)</code> returns a <code>Future</code> ; <code>tell (!)</code> is fire-and-forget   | No                     |
| Akka.NET, Orleans                  | .NET         | thread-pool / turn-based scheduling; grains single-threaded per activation                                     | grain methods return <code>Task&lt;T&gt;</code> ; always asynchronous                           | No                     |
| CAF                                | C++          | cooperative work-stealing scheduler over $P=\#\text{cores}$ workers; <code>detached</code> actors get a thread | <code>request</code> returns a response handle; <code>.then</code> continuation, non-blocking   | No                     |
| SObjectizer                        | C++          | agents bound to dispatchers supplying worker threads/pools                                                     | async by default; <code>so5extra request_reply</code> blocks; runs on provider’s dispatcher     | No                     |
| Actix                              | Rust / Tokio | actors are tasks on Arbiter event-loop threads (M:N)                                                           | <code>send</code> returns a <code>Request</code> future; <code>do_send</code> fire-and-forget   | No                     |
| Pony                               | Pony RT      | work-stealing scheduler, one thread per core; per-actor heaps                                                  | none — behaviours are asynchronous by design                                                    | No                     |
| Swift actors                       | Swift RT     | serial executor per actor over a cooperative pool                                                              | cross-actor access requires <code>await</code> ; synchronous cross-actor calls are prohibited   | No                     |
| Ray, Dapr, distributed Proto.Actor |              | one worker/process or mailbox per actor; M:N pools                                                             | remote call returns a future; turn-based dispatch                                               | No                     |
| <b>fast_send</b> (this work)       | C++/Rust     | per-actor exclusive-access lock; sender’s thread runs the handler                                              | blocking call returns a value on the sender’s thread                                            | <b>Yes</b>             |

microbenchmark reports on the order of 50 million messages per second on a single machine in a ping-pong configuration, a figure the authors show is dominated by that batching setting; it is a vendor microbenchmark, not an application measurement, and should be read as such [21]. The request-reply operator `ask` returns a `Future/CompletionStage` rather than a value and never blocks the caller on the handler; Akka’s documentation explicitly discourages blocking inside actors because it starves the shared pool [7].

**.NET: Orleans and Akka.NET.** Orleans introduced the “virtual actor” (grain) abstraction, in which the runtime activates grains on demand and schedules them turn-based and single-threaded over a shared pool; every grain method is asynchronous and returns a **Task**, so a caller obtains a promise and the grain body runs on the grain’s own activation context, never on the caller’s thread [9]. The Orleans team reported the Halo 4 presence service sustaining roughly 130,000 requests per second across 25 servers with near-linear scaling to 125 servers [22]. A recent independent empirical comparison of Akka, Akka.NET, and Orleans on commodity cloud hardware found Akka.NET and Orleans comparable and both ahead of Akka in a non-distributed setup, with Orleans strongest in the distributed setup [24].

**C++: CAF and SObjectizer.** The C++ Actor Framework schedules cooperatively-run actors over a work-stealing pool of one worker per core, with a lock-free mailbox; actors that must block opt into a **detached** mode that gives them a dedicated thread [8]. Its request-reply operation, **request**, does not occupy a CAF thread while waiting: the requesting actor simply processes no other message until the response arrives, and installs a typed continuation via **.then**; the caller does not execute the callee’s handler [8]. On the Savina cross-language benchmarks [25], a recent study found CAF only marginally ahead of Akka on average (about  $1.09\times$  at 20 threads) and behind it on nine of sixteen benchmarks [16], so the common claim that native C++ actors dominate the JVM is more nuanced than it is often stated. SObjectizer likewise binds agents to dispatchers backed by worker-thread pools and is asynchronous by default; its companion library offers a synchronous **request\_reply**, but the handler still runs on the service provider’s dispatcher, not the caller’s thread.

**Rust: Actix.** Actix [26] runs actors as tasks on Arbiter event loops atop the Tokio runtime, multiplexing many actors per thread. **send** returns a **Request** future that resolves to the handler’s result, and **do\_send** is fire-and-forget; in both cases the handler runs on the target actor’s Arbiter, not the sender’s. (The widely-cited Actix TechEmpower web-framework results concern **actix-web**, not the actor library, and are not evidence about actor-message performance.)

**Pony.** Pony is the clearest case of the mainstream design point. Behaviours are asynchronous by construction — there is no synchronous send at all — so request-reply must be built by hand from callback messages. Pony runs a work-stealing scheduler with one thread per core and gives each actor its own heap, which enables a fully concurrent, per-actor garbage collector (ORCA) that needs neither read nor write barriers, relying instead on reference capabilities for data-race freedom and on causal message delivery for correctness [23]. Its performance comes from cheap actors, local GC, and never blocking.

**Swift.** Swift’s language-level actors (SE-0306) give each actor a serial executor drawn from a cooperative thread pool. Cross-actor access must be **await**-ed and is therefore asynchronous; the compiler *prohibits* synchronous cross-actor calls outright [10]. Swift actors are reentrant: a suspended actor may interleave other work before resuming.

**Distributed runtimes.** Ray, Dapr, and Proto.Actor extend the actor idea across machines. A Ray actor is by default a single worker executing methods one at a time, and a call returns an object reference (a future); Dapr enforces turn-based, single-threaded access to each actor via a per-actor lock and dispatches method calls over HTTP/gRPC; Proto.Actor processes each mailbox

one message at a time on a shared thread-pool dispatcher. In all three the invocation is asynchronous and the sender never runs the receiver’s handler.

**Two structural facts.** Two observations hold across the entire survey. First, on *thread mapping*: the dominant design multiplexes many actors M:N over a small pool of worker or scheduler threads, because assigning a dedicated operating-system thread to each actor does not scale to large actor populations; the few one-thread-per-actor cases (CAF’s **detached** actors, Ray’s threaded actors, process-per-actor deployments) are reserved for blocking or isolation needs and are understood as exceptions. Second, on *synchronous messaging*: where a synchronous or request–reply path exists at all, it is emulated over asynchronous mailbox delivery — the caller either blocks in a receive (Erlang) or, more commonly, receives a future/promise (Akka, Orleans, CAF, Actix, Ray) — and in every case the receiver’s handler runs on the receiver’s own scheduled thread or context. Some systems (Pony, Swift’s cross-actor calls) forbid a synchronous call altogether. Among the runtimes surveyed here we did not find one that executes the receiver’s handler on the sender’s thread under the receiver’s exclusive-access lock. That combination — cross-thread synchronous execution under receiver transparency — is the design point **fast\_send** occupies. It needs protection against synchronous cycles, which Section 7.3 provides for cycles on one thread.

## 4 Why actors? The design case and its cost

Before addressing performance, one question must be settled: why adopt the actor model at all, when a program can already spawn operating-system threads and share memory under mutual exclusion, or drive concurrency through an event loop such as `boost::asio`? Those mechanisms are not deficient in capability. The case for actors is not that they compute something the alternatives cannot; it is what the discipline *removes* — the classes of error it prevents by construction.

**What the discipline removes.** Because each actor owns its state and communicates only by messages, there is no mutable state shared between actors, and therefore no memory-level data race, the most common serious failure of lock-based concurrency. Empirical study of the field supports this: data races and deadlocks are described as “two mistakes that are hard to make with actors” [27], and a comprehensive study of 186 real-world actor bugs confirms that actor programs are markedly less susceptible to low-level data races and deadlocks than multithreaded code, even as they admit their own higher-level bug classes [28, 29]. The programmer reasons about one message at a time against a consistent private state, sequentially, and never writes a lock. The synchronization apparatus that dominates the correctness burden of hand-rolled concurrent code — choosing lock granularity, establishing a global lock order to avoid deadlock, placing memory barriers, and reasoning about every interleaving that a thread scheduler or an `asio` callback chain admits — simply does not appear in an actor program. Fault isolation follows from the same isolation: a failing actor can be restarted by a supervisor without corrupting the state of others, the foundation of the “let it crash” reliability discipline [6].

These properties belong to the asynchronous model. The synchronous delivery of Section 7.1 gives one of them up: a **fast\_send** holds the receiver’s lock while its handler runs, and a chain of **fast\_send** calls holds several actors’ locks at once. A cycle of synchronous calls on one thread is caught by the detector of Section 7.3; a cycle across threads, where thread 1 holds *A* and waits for *B* while thread 2 holds *B* and waits for *A*, deadlocks, exactly as a lock-ordering error does in lock-based code. The asynchronous path keeps the guarantee; the synchronous path trades part of it for latency.

**The cost.** These properties are not free. The message-passing discipline that provides isolation is also the source of the per-message overhead this paper addresses: heap allocation, mailbox queue operations, scheduling, and context switches (Section 9). This overhead is the cost of isolation, and on latency-critical paths it has historically led engineers to abandon the actor model and return to shared memory and locks. The conflict between the model’s safety and its per-message cost is the subject of this paper.

**Determinism and simulation.** A further benefit follows from the same isolation and is especially important in a trading system. Because an actor’s behavior is a function of the message sequence on its mailbox and nothing else — no shared state, and no wall-clock or global-random side channel once the actor obtains time and randomness as messages — an actor is deterministically reproducible: the same input sequence yields the same states, outputs, and faults on every run. This makes the actor a natural unit of *simulation*. When actors are co-scheduled behind one queue (Section 9.1), that single queue is fed by live sources in production or replayed from a recorded or synthetic sequence in simulation, and the actors cannot tell the two apart. The same code is therefore both a real-time execution engine and a deterministic simulator: a trading strategy is developed and backtested against recorded market data, debugged by replaying the exact sequence that produced a fault, and deployed to production without modification. Shared-memory and callback-based designs do not offer this without effort — their behavior depends on thread interleavings and ambient clocks that a simulator must otherwise mock away, whereas for actors the property follows from the message discipline. The duality, and its use in developing an execution algorithm, are the subject of Section 11.

**The generation argument.** A further benefit has become relevant only recently. An actor is a small, regular unit of code: a class with private state and one handler per message type, whose entire observable contract is the set of messages it accepts and the replies it returns. This regularity makes actors unusually amenable to automatic construction by large language model (LLM) code generators, which are fluent in the paradigm: a model can scaffold an actor, write its message handlers, and — because an actor’s behavior is a function from an incoming message to a reply with no hidden shared state — generate self-contained unit tests that drive messages in and assert on the replies, without the elaborate harness that testing shared-memory code requires. The contrast with the alternative is documented. Generating *correct* lock-based concurrent code is hard even for strong models: a dedicated benchmark for concurrent code generation finds that current LLMs struggle to produce correct synchronization and repeatedly introduce concurrency errors [30]. The actor discipline eliminates the hazard that both human authors and code generators handle worst — shared mutable state guarded by hand-written locks — so actor code plausibly is not only faster to produce but more robust and closer to correct than a hand-rolled alternative built on a custom locking scheme. We state this as a design hypothesis, not a result: we do not evaluate generated-code quality here. The supporting evidence is indirect — the demonstrated difficulty of the lock-based alternative [30] together with the measured data-race freedom of the actor approach [27, 28] — and a systematic evaluation of LLM-generated actor implementations, for correctness, concurrency safety, and performance, is left to future work.

**Removing the co-located cost.** Taken together, the benefits of the actor model — data-race freedom, sequential per-actor reasoning, fault isolation, and amenability to automatic generation and testing — have always been offset by a performance penalty that put the model impractical on latency-critical paths. If that penalty is removed for the common case of co-located actors, the

trade-off changes: the isolation, the absence of hand-written locking, and the amenability to code generation are retained, while the per-message cost approaches that of a direct call. Providing that path, safely and without breaking the abstraction, is the purpose of `fast_send`, described next.

## 5 Background and related work

The actor model [1, 2] specifies asynchronous message passing as the sole means of inter-actor communication, and foundational and subsequent treatments present this asynchrony as essential to isolation, fault tolerance, and location transparency [2, 15]. Several systems provide a second, synchronous-looking communication path; each differs from `fast_send` in a way that bears on receiver transparency or on multi-threaded execution.

**Single-threaded synchronous calls.** The E language [3] offers an immediate call `(.)` returning a value and an eventual send `(<-)` returning a promise. E vats are single-threaded, so the immediate call is a conventional method invocation with no lock, no cross-thread execution, and no possibility of deadlock; the model does not extend to actors running on multiple threads. AmbientTalk [4] similarly builds on communicating event loops in which cross-actor communication is always asynchronous. Actor4j [11] optimizes messaging between actors on the *same* thread, and reliable-actor systems such as the Kubernetes Application Runtime (KAR) [12] bypass the queue only for an actor’s calls to itself, not for cross-actor calls.

**Blocking that still uses the receiver’s thread.** Erlang’s `gen_server:call` [5] blocks the caller, but the request still traverses the receiver’s mailbox and is processed on the receiver’s own process; the caller’s thread never executes the receiver’s handler. Akka’s `ask` operator [7] returns a future and remains asynchronous internally, routing through the mailbox, and the framework explicitly discourages blocking inside actors. Orleans [9] presents grain methods in call syntax but is asynchronous underneath (Task-returning), with no caller-thread execution of the grain. Swift’s actors [10] require `await` for cross-actor access and prohibit synchronous cross-actor calls at the compiler level. The C++ Actor Framework (CAF) [8] dispatches handlers on a scheduler’s worker pool, so the caller thread does not run the target’s handler.

**Direct execution without message-passing semantics.** The Active Object pattern [13] interposes a proxy, activation queue, and scheduler so that the client’s thread never runs the servant’s method on a separate thread. The Monitor Object pattern [14] does run a method on the caller’s thread under a lock — sharing the caller-thread-execution mechanism with `fast_send` — but a monitor is a passive object invoked by a direct method call: it has no mailbox, offers no asynchronous alternative, and does not present the callee as an isolated actor that cannot observe how it was reached.

**Deterministic reactor concurrency.** A separate recent line of work constrains, rather than accelerates, actor-style concurrency: Lingua Franca [16] schedules reactors under a logical-time model to make concurrent execution deterministic. That objective is orthogonal to `fast_send`, which addresses the per-message cost of co-located delivery and preserves receiver transparency; the two could be combined, since `fast_send` does not alter the logical ordering of message handling.

## 5.1 Positioning

Across these systems, a synchronous path either (i) is confined to a single thread or same-thread actors (E, AmbientTalk, Actor4j), (ii) blocks the caller while still executing on the receiver’s thread through the mailbox (Erlang, Akka, Orleans, CAF), (iii) is prohibited outright (Swift), or (iv) abandons message-passing semantics for direct method calls on a passive object (Active Object, Monitor Object). What distinguishes `fast_send` is the combination of (a) cross-thread synchronous execution — the sender’s thread runs the receiver’s handler while holding the receiver’s exclusive-access lock — (b) under receiver transparency, with (c) a cyclic-invocation detector that catches synchronous cycles on one thread before they deadlock or overflow the stack; cycles across threads are outside what it can see (Section 7.3). The remainder of the paper develops these three elements.

## 6 Requirements of an HFT data path

Before adapting the actor model, we state what an HFT data path requires of any concurrency substrate it is built on. Two requirements dominate, and together they set the design target for the rest of the paper.

**Performance, specifically the tail.** For an HFT data path performance is the overriding requirement, and the figure of merit is not the mean latency but the tail — the 99th, 99.9th, and 99.99th percentiles. Market data decoded late and orders sent late are worthless or produce a loss, so a strategy is judged on its worst-case reaction time, not its typical one. A concurrency layer that is fast on average but occasionally slow is therefore unacceptable in a way that has no analogue in throughput-oriented systems.

**Arrivals are clustered, so tails are intrinsic.** Market data does not arrive smoothly. Market events arrive in bursts rather than evenly — the feed is strongly non-Poisson, as ours confirms (Section 10.7) — and the exchange coalesces the events available at each send into a single datagram, so a burst of events becomes a long packet. A long packet converts into a long latency tail, because its messages are decoded in sequence. A path designed for the *average* arrival rate would therefore be mis-designed; the system must stay fast enough under a burst that its servicing queue does not build, which makes the quantities to engineer the per-message service time and its tail, not the mean throughput.

**Consequence for the concurrency layer.** The two requirements combine into a single target for the messaging substrate: it must add negligible per-message cost *and no latency tail of its own*, while preserving the actor model’s isolation and single-message reasoning (Section 4). The four changes of the next section are built to meet this target, and the case study of Section 10 then confirms the per-hop cost is small: with decode and book update on one thread the book-update median is about 1  $\mu$ s, of which a `fast_send` hop is about 1%. It also shows the other side of the same argument: when each datagram instead crosses a thread through an asynchronous mailbox, the median is about 7  $\mu$ s and part of the tail is mailbox-queue backlog, so where the path crosses threads is a design decision in its own right.

## 7 Adapting the actor model for HFT

The survey of Section 3 fixed the mainstream design point: actor runtimes deliver every message asynchronously, give each actor its own mailbox, and multiplex many actors M:N over a small pool of worker threads. On that design even a request–reply between two actors in the same process pays a heap allocation, an enqueue and a dequeue, a scheduler dispatch, and a context switch — and, for the reply, all of it again. That is the correct design *across machines*, where the asynchrony buys isolation, fault tolerance, and location transparency (Section 4); it is over-provisioned for actors that share an address space, where the sender’s own thread could run the receiver’s handler directly.

To meet the requirements of Section 6, we make four changes to that design. Each is a departure from one standard-actor assumption, and each removes a specific per-message cost — or its tail — while preserving the isolation, sequential per-actor reasoning, and the receiver transparency that motivate the model (Table 2). The first and central change, **fast\_send**, occupies Sections 7.1–7.4; the remaining three — actor groups, selectable mailbox queues, and a memory pool — follow in Sections 7.5–7.7.

**Table 2.** The four changes as departures from the mainstream actor design. Each removes a co-located per-message cost; none alters the isolation or the message-passing semantics a handler observes.

| Dimension      | Mainstream actor model                                                        | This work (kaspar-hft)                                                                                                          |
|----------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Delivery       | asynchronous only; the handler runs later on the receiver’s scheduled thread  | synchronous inline delivery on the sender’s thread ( <b>fast_send</b> ) for co-located actors; asynchronous retained for remote |
| Actor → thread | one mailbox per actor, M:N over a worker pool, one scheduled turn per message | a <i>group</i> co-schedules actors on one thread behind one shared mailbox                                                      |
| Mailbox queue  | one fixed queue implementation                                                | four implementations, selectable per actor to its write pattern                                                                 |
| Message memory | heap-allocated per message                                                    | stack-allocated for synchronous requests; pooled when it must outlive the call                                                  |

### 7.1 Synchronous delivery: **fast\_send**

Each actor has a message handler, a mailbox, private state, and a synchronization mechanism that guarantees at most one thread executes its handler at a time (a mutex by default; a lock-free primitive is admissible). Two delivery operations share this structure.

**Asynchronous delivery (**send**).** The sender heap-allocates the message, enqueues it on the receiver’s mailbox under the mailbox’s synchronization, and returns immediately with no value. Later, a thread assigned to the receiver dequeues the message, acquires the receiver’s exclusive-access lock, and runs the handler. A reply, if any, is heap-allocated, wrapped in a response message, and enqueued on the sender’s mailbox.

**Synchronous delivery (**fast\_send**).** The sender may allocate the message on its *stack*, because the sending thread blocks until processing completes and the message’s lifetime is therefore bounded by the call. The sender’s thread acquires the *receiver’s* exclusive-access lock (blocking if another

thread holds it), pushes the receiver onto a thread-local call chain (Section 7.3), and invokes the receiver’s handler directly on the sender’s thread. The handler runs its identical code and optionally calls the reply operation; on return, the receiver is popped from the call chain, the lock is released, and `fast_send` returns the reply value (or null if none was set). A stack-allocated message is reclaimed when the sender’s frame unwinds.

In both modes the receiver’s exclusive-access lock is held for the duration of handler execution, so the “one message at a time” guarantee holds identically under the conservative detection policy; the permissive (actor, message-class) refinement of Section 7.3 admits same-actor re-entrancy and relaxes it. The distinction from a naive direct call is that the synchronous path holds the receiver’s exclusive-access lock and routes the reply through the same operation as the asynchronous path; it is delivery of a message that happens to execute on the sender’s thread.

**Priority semantics.** A `fast_send` invocation does *not* join the receiver’s mailbox behind messages already queued there. It executes as soon as it acquires the receiver’s exclusive-access lock, so synchronous delivery has *priority* over queued asynchronous delivery: if the receiver’s mailbox holds  $m_1, m_2, m_3$  and a sender issues `fast_send( $m_4$ )`, then once the lock is free  $m_4$ ’s handler runs first, and  $m_1, m_2, m_3$  run afterward on the receiver’s own turn. This is intentional. `fast_send` is a high-priority, synchronous request/response for a caller that needs the receiver’s current state without waiting for the mailbox to drain; it is not a faster spelling of an ordinary asynchronous `send`, and it therefore reorders that one invocation ahead of the queue. The exclusive-access lock is still respected: if the receiver is mid-handler when `fast_send( $m_4$ )` arrives, the sender blocks on the lock and  $m_4$  runs after that handler returns but still ahead of the remaining queue. A caller that requires strict FIFO with respect to other senders must use `send`.

**Unified reply routing.** A single reply operation serves both modes and selects its behavior from delivery-context information rather than from any argument the handler supplies. Under asynchronous delivery it heap-allocates the value, wraps it in a response message, and enqueues it on the sender’s mailbox; under synchronous delivery it heap-allocates the value and stores it in a return-value location that the blocked `fast_send` reads and returns. The value is heap-allocated in *both* modes deliberately: if the synchronous path could stack-allocate the reply while the asynchronous path could not, the handler would have to know the mode, defeating transparency. The reply may be invoked at most once per message.<sup>3</sup> An opt-in optimized reply stores the value without heap allocation for code paths known to be synchronous-only;<sup>4</sup> it raises an error if the message was delivered asynchronously, and by design breaks transparency, so its use is explicit.

**Chained synchronous execution.** If actor  $A$  synchronously invokes  $B$ , which synchronously invokes  $C$ , and so on, the entire chain executes on  $A$ ’s thread, which holds the exclusive-access locks of all actors in the chain simultaneously and, when those locks are uncontended, performs no context switches. Such an  $N$ -actor synchronous chain incurs  $O(1)$  context switches, against  $O(N)$  for the equivalent asynchronous chain, and executes on a single core, avoiding the cross-core cache-line migration of state that the asynchronous path incurs. A contended lock in the chain blocks its thread, which reintroduces a context switch at that step.

---

<sup>3</sup>kaspar-hft does not enforce this yet: a second reply to a synchronous delivery replaces the first, and a second reply to an asynchronous one sends a second message.

<sup>4</sup>kaspar-hft does not implement the optimized reply yet.

## 7.2 Receiver transparency

Receiver transparency is the property that the receiving actor’s handler is written identically for both delivery modes and does not depend on (i) whether it was reached synchronously or asynchronously, (ii) which thread is executing it, or (iii) whether the sender is blocked awaiting a reply. It is a property of how handlers are written, not isolation that the framework or the language enforces. A handler that looks can infer the mode: C++ cannot stop it from inspecting its own execution context — its thread identifier, the address of a local variable (and so which thread’s stack it is on), or the time — and some message fields the framework sets are filled differently in the two modes (element 4). Transparency means that a handler has no reason to look, and that its code does not change between modes. It is what keeps `fast_send` within the actor abstraction rather than reducing it to a locked method call.

Four elements bear on it:

1. **Identical interface.** The handler receives a message and invokes the same reply operation in both modes; there is no mode flag, and no conditional branch in the handler depends on the delivery mode.
2. **Exclusive access in both modes.** The receiver’s exclusive-access lock is held throughout handler execution whether the handler runs on the receiver’s own thread (asynchronous) or on the sender’s thread (synchronous), so the observable execution environment — single-writer, sequential — is the same.
3. **Mode-independent reply and allocation.** The unified reply routes by delivery context with no handler involvement, and the reply value is heap-allocated in both modes, so neither the reply’s behavior nor allocation side effects reveal the mode.
4. **No delivery flag for handlers.** The flag recording how a message was delivered is private to the framework. Other message fields are not mode-neutral: the queue depth and the “mailbox empty” flag describe the receiver’s mailbox on asynchronous delivery and have fixed values on synchronous delivery, and the destination field is set only on asynchronous delivery. They exist for instrumentation and routing; a handler whose logic depends on them gives up transparency.

Two consequences follow. First, the delivery mode becomes a *sender-side* optimization: a sender may upgrade an asynchronous send to a synchronous one with no change, recompilation, or redeployment of the receiver. Second, the mechanism is distinct from a direct procedure call, in which the callee is a passive object invoked on the caller’s stack with no mailbox and no exclusive-execution guarantee; here the receiver remains an isolated actor that perceives the arrival of a message and is unaware that the sender’s thread is the one running it.

## 7.3 Cyclic-invocation detection

**What the detector cannot see: cycles across threads.** The call chain below is per thread, so it detects only a cycle that stays on one thread. It cannot see a cycle spread across threads: thread 1 runs a `fast_send` to *A* whose handler does a `fast_send` to *B*, while thread 2 runs a `fast_send` to *B* whose handler does a `fast_send` to *A*. Each thread’s check passes —  $B \notin \{A\}$  and  $A \notin \{B\}$  — and each thread then blocks on the lock the other holds. `fast_send` is callable from any thread, so this case is real. Catching it needs shared state rather than a thread-local chain: a global wait-for graph checked before blocking, a global order on actor locks, or a bounded try-lock that falls back

---

**Algorithm 1:** Synchronous send with cyclic-invocation detection

---

**thread-local:** call chain  $C$  (set of actors executing synchronously on this thread)

**Input:** target actor  $B$ , message  $m$

```
1 if  $B \in C$  then
2   return cyclic-invocation signal (exception or error indicator) // before any lock is
   taken
3 acquire  $B$ 's exclusive-access lock
4  $C \leftarrow C \cup \{B\}$ 
5  $r \leftarrow B.\text{handler}(m)$  // runs on this (the sender's) thread
6  $C \leftarrow C \setminus \{B\}$ 
7 release  $B$ 's exclusive-access lock
8 return  $r$ 
```

---

to asynchronous delivery (the `fast_send_void` policy of Section 7.4). Choosing among these is future work (Section 13).

Synchronous invocation introduces a failure mode that asynchronous delivery does not have. If synchronous calls form a cycle —  $A \rightarrow B \rightarrow C \rightarrow A$  — then under a non-reentrant lock the thread waits to acquire an actor's lock it already holds, and the system deadlocks; under a reentrant lock the handler re-enters with the lock's recursion count incremented, producing unbounded nested execution that overflows the stack and exposes inconsistent intermediate state. Neither is acceptable, and the cycle cannot be discovered after the offending lock is taken — by then the thread is already stuck.

### 7.3.1 Call-chain membership test

The mechanism maintains, per thread, a *call chain*<sup>5</sup>: the ordered set of actors whose handlers are currently executing on that thread via synchronous invocation. Because it is thread-local, it requires no synchronization and resets automatically between tasks. It is initialized when a thread begins processing a dequeued (asynchronous) message — the dequeued actor is its first entry — and each synchronous invocation pushes its target on entry and pops it on completion. The chain may be represented as a stack or list, a hash set for  $O(1)$  membership queries, a bitmap indexed by actor identifier, or a hybrid.

Before acquiring the target's lock, a synchronous send queries whether the target is already present in the chain. If it is not, the send proceeds normally; if it is, a cyclic invocation has been detected, and the mechanism signals it — by raising a cyclic-invocation exception carrying the target identity and the full chain, or by returning an error indicator — *without* acquiring the lock or invoking the handler. Algorithm 1 states the check. Because the query precedes lock acquisition and reads only thread-local state, its overhead is negligible against the cost of a lock acquisition and a handler invocation, and the chain depth is small in practice.

### 7.3.2 Refined detection with message classes

Membership keyed on actor identity alone is conservative: it rejects legitimate callbacks in which  $A$  synchronously invokes  $B$  and  $B$  synchronously invokes a *different* handler of  $A$ . A refined

---

<sup>5</sup>kaspar-hft does not implement the call chain yet.

mode<sup>6</sup> keys the call chain on (actor, message-class) pairs, where the message-class discriminator may be a type tag, a class identifier, a handler function pointer, or any code-path discriminator. A cycle is signaled only when the same pair recurs. This admits a callback pattern such as  $A/\text{compute} \rightarrow B/\text{validate} \rightarrow A/\text{validated}$  — three distinct pairs — while still catching genuine unbounded recursion such as  $A/\text{foo} \rightarrow B/\text{bar} \rightarrow A/\text{foo}$ , in which a pair repeats. Because permissive keying allows an actor’s state to be re-entered by a nested handler, reentrancy safety in that mode is the programmer’s responsibility. The policy — conservative (actor-only) or permissive (pair-based) — is configurable system-wide, per-actor, or per-invocation.

## 7.4 A family of delivery operations

The same transparent handler interface supports five delivery operations<sup>7</sup> that differ only in how the sender behaves when the receiver’s lock is contended or a cycle is detected (Table 3). All synchronous variants perform the cyclic-invocation check of Section 7.3 before attempting the lock.

**Table 3.** The five delivery operations. All share one receiver-transparent handler interface; they differ in sender-side behavior under contention and cycles.

| Operation                              | Blocks | Returns | Lock unavailable                 | Cycle detected |
|----------------------------------------|--------|---------|----------------------------------|----------------|
| <code>send</code> (async)              | no     | —       | (enqueues)                       | —              |
| <code>fast_send</code> (blocking sync) | yes    | value   | waits (sleep on lock)            | exception      |
| <code>fast_send_spin</code>            | yes    | value   | waits (busy-wait)                | exception      |
| <code>fast_send_x</code>               | no     | value   | <code>ActorBusy</code> exception | exception      |
| <code>fast_send_void</code>            | no     | —       | async fallback                   | async fallback |

- **fast\_send** blocks on the receiver’s lock by sleeping, and raises a cyclic-invocation exception on a cycle. It is the default synchronous operation.
- **fast\_send\_spin** is identical except that it busy-waits on the lock, optionally issuing a CPU `pause` instruction between attempts, trading CPU cycles for lower wake latency by avoiding the scheduler. It suits short critical sections on latency-sensitive paths with spare cores.
- **fast\_send\_x** attempts the lock without blocking: on success it runs the handler and returns the value; on contention it raises an `ActorBusy` exception, distinct from the cyclic-invocation exception, so the caller can fall back, retry, or propagate. Its execution time is bounded.
- **fast\_send\_void** attempts the lock without blocking and returns no value: on success it runs the handler; on contention *or* on a detected cycle it enqueues the message asynchronously and returns, copying a stack-allocated message to the heap first if necessary. Because it never blocks on the target lock, the operation itself cannot be the waiting party in a lock-wait deadlock; the handler it runs on success may still block or recurse. It fits notification, forwarding, pipeline, and broadcast patterns. The ratio of synchronous completions to asynchronous fallbacks is a useful contention metric.
- An adaptive variant spins for a bounded count and then transitions to blocking acquisition, combining low latency under light contention with bounded CPU use under heavy contention.

<sup>6</sup>kaspar-hft does not implement the refined mode yet.

<sup>7</sup>kaspar-hft implements `send` and `fast_send`; it does not implement the other three, or the adaptive variant, yet.

Selection among the operations is a sender-side decision governed by whether the caller needs a return value, can tolerate an unbounded wait, and has idle processor capacity available; the receiver’s handler is unaffected in every case.

## 7.5 Actor groups

A *group* co-schedules a set of actors on a single thread behind one shared mailbox: the members do not own separate queues, and an asynchronous `send` between two actors in the same group never crosses a core — it is a push and a pop on a queue owned by the running thread, with no cross-thread wakeup. This is the standard low-latency actor optimization (same-thread messaging in Actor4j, a shared dispatcher in Akka; Section 3), and it is the substrate `fast_send` builds on: grouping the actors of a servicing stage onto one thread lets an  $N$ -actor pipeline collapse to a single inline `fast_send` call chain (Section 8). The single shared queue has a second consequence beyond latency — it imposes one total order on every message entering the group, which is what lets the same actor code run unchanged in live trading and in deterministic backtest (Section 11). Section 9.1 reports the grouped round-trip cost.

## 7.6 Selectable mailbox queues

An actor’s mailbox is not a fixed data structure. It is a `std::variant` over four multi-producer/single-consumer queue implementations, chosen per actor to match its write pattern: `BQueue` (a mutex and condition variable around a ring buffer), `BQueueBatched` (the same lock, but the consumer drains the whole queue under one acquisition), `ShardedBQueue` ( $N$  independently locked lanes joined by an atomic cursor), and `LockFreeMPSC` (a bounded lock-free Vyukov ring, in which a producer reserves a slot with one atomic operation). Because the variant holds the concrete type, `push` and the drain loop are devirtualized and inlined. No one implementation is fastest in every regime, so the choice is exposed rather than fixed; Section 9.2 measures the four and gives the selection rule.

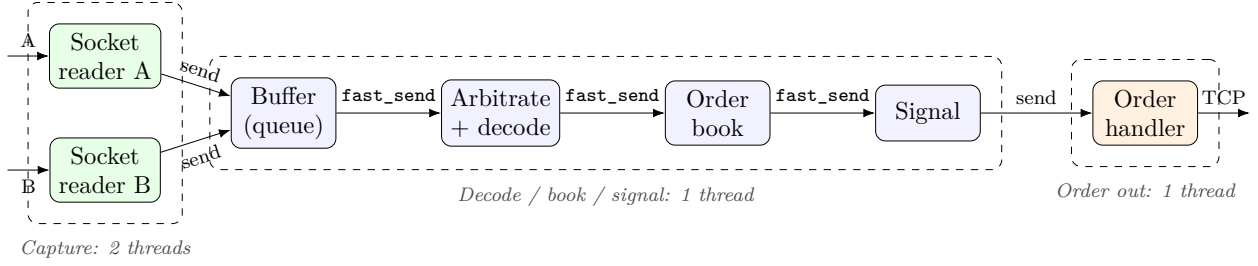
## 7.7 Memory pool

The message-passing discipline heap-allocates a message per interaction, and on a hot path that allocation is both a per-message cost and, worse, a source of tail latency when the general allocator takes its occasional slow path. Two mechanisms remove it. A synchronous `fast_send` request needs no allocation at all: the sender blocks until the handler returns, so the message can live on the sender’s stack (Section 7.1). A message that must outlive the call is drawn from a fixed-size `MemoryPool` rather than `new/delete`, which lowers the mean cost and, more importantly, removes the allocator’s multi-millisecond tail. Section 9.1 measures both effects.

# 8 A representative HFT system architecture with `fast_send`

This section shows where the four changes sit in a real market-data-to-order pipeline, and how the servicing chain is kept fast enough to hold its queue near empty under the clustered arrivals of Section 6; the case study of Section 10 then measures what the framework contributes on a live feed.

Figure 1 shows the thread architecture of a representative HFT market-data-to-order pipeline, drawn from the kaspar-hft reference implementation. CME’s MDP 3.0 market data arrives as two redundant multicast feeds, channel A and channel B. The system decomposes into three sections connected by exactly two cross-thread hops.



**Figure 1.** Thread architecture of a representative HFT pipeline. The only cross-thread hops on the critical path are the two asynchronous `sends` (reader→buffer and signal→order-handler); each such hop is the expensive operation, measured at  $\sim 3.4 \mu\text{s}$  per round trip (Table 5), because it wakes another thread. The entire decode/book/signal servicing chain instead collapses onto a single thread via `fast_send`, each hop costing  $\sim 9 \text{ ns}$  — about one mispredicted virtual call (Table 6), three orders of magnitude less than a thread-crossing `send` (see Table 5) — so it incurs no queue and no context switch. The per-message service time of that servicing chain, and hence how fast its queue drains, is set by the aggregate cost of these `fast_sends`. The case study (Section 10) measures the capture-to-book part of this path both with the reader decoding and updating the book on its own thread (book-update medians  $0.8\text{--}1.1 \mu\text{s}$ ) and with the reader→buffer hand-off shown here (median  $\sim 7 \mu\text{s}$ ).

**Section 1 — capture (two threads).** One socket-reader actor per channel does nothing but drain its UDP socket and hand each datagram to the next stage by an *asynchronous send*. It must be asynchronous: a reader that made a synchronous, blocking call would stop draining its socket while the callee ran and would drop packets under a burst — defeating the reader’s only purpose. In *kaspar-hft* the reader pre-allocates message buffers and batches whatever the socket has available before sending, and its handoff to the buffer stage is an *async send*, never a `fast_send`.

**Section 2 — decode, book, and signal (one thread).** The datagrams land in a buffer actor — the queue — whose service handler forwards each message down the remainder of the chain by `fast_send`: arbitration between the A and B feeds, Simple Binary Encoding decode, order-book construction, and signal generation all run synchronously, in sequence, on this one thread, with no intervening queue and no context switch between stages. This is the chained synchronous execution of Section 7.1: an  $N$ -stage chain with  $O(1)$  rather than  $O(N)$  context switches. This is the queue-servicing code whose aggregate per-message time determines how fast the market-data queue drains.

**Section 3 — order out (one thread).** When the signal decides to act, it hands an order to an order-management actor by an asynchronous `send`; that actor performs the TCP/iLink write. This hop is asynchronous by design: the network write takes several microseconds, and blocking the decode-and-book thread on it would stall servicing of the market-data queue and let it back up — the exact failure the architecture exists to avoid.

**Two cross-thread handoffs.** On the critical path from wire to order there are just two cross-thread hops — reader→buffer and signal→order-out. Each is an asynchronous `send` that may wake another thread, and when it does the wakeup and any attendant context switch cost a few microseconds — a thread-to-thread switch measures near  $1.2\text{--}1.5 \mu\text{s}$  on a pinned core, rising to a few microseconds once cache-working-set effects are included [31, 33, 32]. Everything between them is collapsed onto a single thread by `fast_send`. The goal is not to remove those two handoffs; they are cheap and necessary, because the capture and transmit stages must not block the servicing stage.

The goal is to make the servicing stage between them fast enough that its queue never grows, which is what `fast_send` does by collapsing the chain onto one thread. The mechanism’s contribution is thus best measured not as a per-call latency but as the aggregate per-message service time it sustains on the servicing chain.

## 9 Performance analysis

The synchronous path removes, rather than optimizes, the per-message operations of the asynchronous path. Table 4 summarizes the operations eliminated for a co-located delivery.

**Table 4.** Per-message operations on the asynchronous path and their status on the synchronous path, for co-located actors. This is an analytical account of work removed.

| Per-message operation              | Asynchronous          | Synchronous                       |
|------------------------------------|-----------------------|-----------------------------------|
| Message heap allocations           | 1–3                   | 0 request (stack-alloc.), 1 reply |
| Mailbox queue operations           | 2 (enqueue + dequeue) | 0                                 |
| Scheduler invocations              | 1–2                   | 0                                 |
| Context switches (request/reply)   | 2–4                   | 0                                 |
| Context switches, $N$ -actor chain | $O(N)$                | $O(1)$                            |

Each eliminated item has a concrete cost. A context switch is on the order of  $10^3$ – $10^4$  processor cycles and evicts cache and TLB state [31, 32, 33]. Heap allocation of short-lived messages produces fragmentation and, in managed runtimes, collection pauses; stack allocation reclaims the message on frame unwind with no allocator or collector involvement. Removing the mailbox operations removes their synchronization traffic on the memory bus; quantifying that reduction requires measurement and is left to future work. The chained-execution result —  $O(1)$  rather than  $O(N)$  context switches across an  $N$ -actor synchronous chain — is the largest structural effect for deep call graphs.

**Context: measured cost of the asynchronous path.** To indicate the magnitude of the overhead the synchronous path targets, we measured the asynchronous messaging path of the same actor runtime at 5.5 to 13.6 million round trips per second after successive batching and memory-pooling optimizations; in the saturated throughput regime a message costs on the order of 74 ns, while in the sparse regime an individual send costs several microseconds (4–7  $\mu$ s), with condition-variable wakeups costing 1–3  $\mu$ s each. These are measurements of the *asynchronous* path and bound the per-message costs that synchronous delivery removes; they are not measurements of `fast_send`.

**Batching the asynchronous path.** The  $2.46\times$  batching-and-pooling gain above is worth a note, because it is the cross-thread counterpart of `fast_send`: where `fast_send` removes the mailbox from the co-located same-thread path, batching amortizes it on the cross-thread path that must stay asynchronous (the two capture and transmit hops of Figure 1). The baseline pays a mailbox-lock acquisition on every push and pop, an actor-lock acquisition per dispatch, and a condition-variable wakeup per send. Draining the whole queued burst under a *single* mailbox-lock acquisition, and signaling the condition variable only on the empty-to-non-empty transition, raised throughput from 5.5 to 8.96 million round trips per second ( $1.63\times$ ); adding the memory pool reached 13.6 million ( $\sim 74$  ns per message). This is the `BQueueBatched` mailbox of Section 9.2. The gain is regime-dependent in exactly the way the workload requires: it is negligible on a near-empty queue and maximal when a burst has backed the queue up — the same clustered-arrival regime

(Sections 6, 10.7) that produces the latency tail — so batch draining is what holds the asynchronous stage fast under precisely the load that matters. One result runs counter to intuition and fixes where the batching must live: batching on the *sender* side instead — pushing many messages under one lock — *lowered* throughput to 11.3 million, because it serialized producer and consumer and destroyed their pipeline overlap; fewer lock acquisitions did not compensate. Batching therefore belongs on the drain side only.

We turn now to direct microbenchmarks of the synchronous path itself.

## 9.1 Microbenchmarks of the C++ implementation

Table 4 predicts that eliminating each per-message operation produces a measurable reduction in latency. The framework’s C++ microbenchmark suite measures these steps on a *sequential* round trip — ping → pong → reply with a single message in flight, so each sample is a latency rather than a saturated throughput [34]. The figures below were taken on an AMD EPYC 9374F (RHEL 9, g++ 15), compiled `-O3 -march=native` and pinned with `taskset` to two cores, reported as the median of three runs. Absolute values depend on hardware, allocator, thread placement, and turbo state; the reproducible result is the *ratio* between rows, each of which isolates one eliminated operation. The `steady_clock` resolution on this machine quantizes to  $\sim 10$  ns, so a per-sample `fast_send` percentile is resolution-bound; `fast_send` costs are therefore also reported as amortized ns (total wall time over  $N$  operations, a single clock-read pair), which resolves costs below one tick.

Table 5 isolates the cost of crossing threads; the two paragraphs that follow it decompose the internal overhead of `fast_send` and the effect of the memory pool.

**The cost is thread-crossing, not messaging.** Table 5 holds the work constant — one ping/pong round trip with a reply — and varies only the delivery mechanism, so each row adds back one operation. A cross-thread asynchronous `send` between two actors on separate cores costs  $\approx 3.4 \mu\text{s}$  at the median. This cost is dominated by the two cross-core mailbox wakeups per round trip, each a mutex acquisition and a condition-variable signal on which the receiving thread must be scheduled; it is placement-sensitive, because the cross-core path depends on which cores the two threads occupy. Keeping both threads but letting each busy-poll its mailbox instead of sleeping removes the wakeups and nothing else: the round trip falls to 370 ns. Of the  $\sim 3.4 \mu\text{s}$ , therefore, some  $3 \mu\text{s}$  is putting threads to sleep and waking them, and about 370 ns — two crossings of  $\sim 185$  ns — is moving the message and the queue slot between cores; the price of spinning is a core per actor, busy whether or not a message arrives. Co-scheduling the two actors on one thread (grouping) removes the crossings as well and reduces the round trip to 90 ns, a  $37\times$  reduction. Replacing the remaining mailbox push/pop with an inline `fast_send` reduces the round trip to  $\sim 30$  ns (amortized; the per-sample  $p50$  is resolution-bound), because a `fast_send` is mechanically a locked function call: it takes the receiver’s lock and runs the handler inline. This  $\sim 30$  ns is the whole round trip — the ping and pong invocations plus the reply. What that costs against the dispatch a programmer would otherwise write by hand is measured separately, in Table 6.

**Against hand-written dispatch.** Table 5 compares actor delivery paths with each other. The comparison that decides whether to use the framework at all is against the call a programmer would otherwise write. Table 6 measures that on the same machine: every arm does the same work on a stack-allocated message and differs only in how the work is reached. It measures dispatch alone, without a reply, so its `fast_send` figure is below the round trip of Table 5. Direct and virtual calls are real, non-inlined calls. A virtual call costs very little when the branch predictor knows its target

**Table 5. The cost is thread-crossing, not messaging.** Same ping/pong round trip with a reply; only the delivery mechanism varies. Each row eliminates one operation relative to the row above. “Wakeup” is a cross-core mailbox wakeup of a sleeping thread (mutex + condition variable); “queue” is a mailbox push/pop. The *p50* column is per-sample round-trip latency; the **fast\_send** figure is amortized, because its per-sample *p50* is resolution-bound (the **steady\_clock** tick is  $\sim 10$  ns), and its speedup is computed on amortized figures. The grouped and **fast\_send** rows show negligible spread. In the spinning row both actors keep their own threads on separate cores, but each thread busy-polls its mailbox (the lock-free mailbox with a consumer that never sleeps), so a message still crosses cores but never waits for a thread to wake. Three runs: *p50* 370–380 ns (median 370), *p99* 400–410 ns.

| Delivery mode                     | Threads | Wakeup | Queue | <i>p50</i> RT (ns) | Speedup          |
|-----------------------------------|---------|--------|-------|--------------------|------------------|
| <b>send</b> , ungrouped           | 2       | yes    | yes   | $\sim 3370$        | $1\times$        |
| <b>send</b> , ungrouped, spinning | 2       | no     | yes   | 370                | $9.1\times$      |
| <b>send</b> , grouped             | 1       | no     | yes   | 90                 | $37\times$       |
| <b>fast_send</b> (inline)         | 1       | no     | no    | $\sim 30$          | $\sim 110\times$ |

and several nanoseconds when it does not, so the table reports both: a single “virtual call costs  $X$ ” figure would depend entirely on how predictable the call site is.

**Table 6. fast\_send against hand-written dispatch.** Same work in every arm; amortized ns per call, minimum of 15 runs of  $5 \times 10^6$  calls, AMD EPYC 9374F, RHEL 9.2, `g++ -O3 -march=native`, pinned to two cores. Dispatch only, no reply. Source: `actors/cpp/perf/bench_dispatch.cpp`.

| Dispatch                                 | ns        | $\times$ direct |
|------------------------------------------|-----------|-----------------|
| direct call (not inlined)                | 1.63      | 1.0             |
| virtual call, one target (predicted)     | 1.86      | 1.1             |
| virtual call, 2–8 targets (mispredicted) | 7.53–9.32 | 4.6–5.7         |
| <b>fast_send</b>                         | 9.10      | 5.6             |

**fast\_send** costs about one virtual call whose target the predictor cannot guess, and about  $5\times$  a predicted virtual call or a direct call: some 7.5 ns more per hop, for the receiver’s lock, message-field setup, handler lookup and reply plumbing. Whether that is small depends on what it is compared with. Against the  $\sim 1\ \mu\text{s}$  one-thread socket-to-book median of Section 10 it is about 1%; on a leaner path — a hardware-timestamped, wire-to-book pipeline measured in hundreds of nanoseconds — the same 7.5 ns per hop would matter.

**Grouping, and why fast\_send improves on it.** The grouped row is the strongest conventional alternative. Grouping (Section 7.5) co-schedules the two actors on one thread behind one shared mailbox, removing both cross-core wakeups and reducing the round trip from  $\sim 3.4\ \mu\text{s}$  to 90 ns. Grouping still pays for the queue: the message is enqueued, the run loop dequeues it, and the handler runs on a later turn. **fast\_send** removes that layer. On the same thread it does not enqueue; it takes the receiver’s lock and runs the handler inline, returning the reply as a value, in  $\sim 30$  ns — about  $3\times$  faster than grouping. Grouping removes the thread hop; **fast\_send** removes the thread hop *and* the queue. This synchronous, inline, receiver-transparent delivery is, to our knowledge, absent from the mainstream actor systems surveyed in Section 3. The intended use follows: group the actors of a servicing stage onto one thread and drive them with **fast\_send**, so that an  $N$ -actor pipeline collapses to a single inline call chain (the middle section of Figure 1).

**Allocation, not dispatch, is the only variable cost.** Beyond the inline dispatch, the only per-message cost a caller pays is for memory it chooses to allocate. A request message kept on the stack requires none, because `fast_send` blocks until the handler returns (Section 7.1). For a message that must outlive the call, allocation from the `MemoryPool` of Section 7.7 replaces a general-purpose `new/delete`. On a grouped-`send` round trip with two allocations, the pool reduces the amortized cost from  $\sim 128$  ns (pool off) to  $\sim 92$  ns (pool on). The larger and platform-independent benefit is the tail: in a separate run the maximum round trip fell from 5.5 ms to  $33\ \mu\text{s}$  once pooled, because the pool never takes the general allocator’s occasional slow path: a two-order-of-magnitude reduction of the  $p99.9+$  tail that an unpooled servicing stage would otherwise inject into the path.

**Scope.** These are sequential latencies with one message in flight, not saturated throughput, and they do not vary contention or chain depth. The established results are: `fast_send` costs tens of nanoseconds; its overhead over a direct call is  $\sim 7.5$  ns plus whatever memory the caller allocates; it improves on same-thread grouping by about  $3\times$ ; and pooling removes the latency tail. A saturated, multi-axis evaluation remains future work.

## 9.2 Mailbox queue types and their selection

The four queue implementations of Section 7.6 — `BQueue`, `BQueueBatched`, `ShardedBQueue`, and `LockFreeMPSC` — separate on three regimes, which Table 7 measures on the EPYC target. A fourth regime, a single cross-thread message, distinguishes none of them, because thread placement moves the figure by  $2.2\times$  — more than the gap between queues.

**Table 7. Four mailbox queue types, one column per distinguishing regime** (EPYC target, pinned, median of three runs). *Grouped*: single-thread round-trip median. *Burst*: single bursty producer, amortized cost per message. *Fan-in*: 32 concurrent producers, push  $p99$ . Bold marks the best in each regime; no type is best in all three.

| Queue type                 | Grouped $p50$ (ns) | Burst (ns/msg) | Fan-in $p99$ ( $\mu\text{s}$ ) |
|----------------------------|--------------------|----------------|--------------------------------|
| <code>BQueue</code>        | 90                 | 605.8          | 69.9                           |
| <code>BQueueBatched</code> | 90                 | <b>584.9</b>   | 72.4                           |
| <code>ShardedBQueue</code> | 190                | 983.1          | <b>6.6</b>                     |
| <code>LockFreeMPSC</code>  | <b>70</b>          | 672.0          | 16.5                           |

No single queue wins everywhere. On a shared (grouped) thread, `LockFreeMPSC` has the lowest median (70 ns,  $1.3\times$  below `BQueue`), because its push and pop are each an uncontended atomic operation with no lock or wakeup. Under a bursty single producer — the market-data regime of the case study — `BQueueBatched` is fastest (584.9 ns/msg): draining an entire packet burst under one lock acquisition amortizes the lock over the burst, the drain-side batching whose throughput effect is decomposed in Section 9.1, and `BQueue` trails by under 4%. Under 32-producer fan-in, `ShardedBQueue` wins decisively —  $p99$  of  $6.6\ \mu\text{s}$  against `BQueue`’s  $69.9\ \mu\text{s}$  ( $10.6\times$ ), and  $3.6\times$  the throughput — because its independent lanes remove the single-mutex serialization that dominates the other three; it pays for those lanes in the other regimes, where it is the slowest of the four.

A selection heuristic for the tested EPYC configuration follows; it is a measured guideline for this hardware and these regimes, not a general rule. `BQueue` is a reasonable default: it is competitive in the grouped and bursty regimes (within  $\sim 30\%$  of `LockFreeMPSC` grouped, within 4% of `BQueueBatched` bursty) and has the most stable tail of the four (grouped  $\max$  2.4–9  $\mu\text{s}$ , versus 14–16  $\mu\text{s}$  for `LockFreeMPSC`). `BQueueBatched` is preferable for bursty single-producer mailboxes such as a market-data ingest, and `ShardedBQueue` for the few actors written by many producers

concurrently — a shared order-out actor or a fan-in aggregator — the one regime in which it wins decisively. `LockFreeMPSC` has the best grouped median but the least stable tail measured, so we do not default to it; and `ShardedBQueue`’s lanes are wasted, and counterproductive, outside fan-in.

## 10 Case study: tick-to-book latency in a live market-data path

The microbenchmarks of Section 9.1 measure the framework’s per-message cost in isolation. This section grounds that number in a live measurement of the KASPAR-HFT market-data path on production exchange data, and asks the paper’s title question directly: is the actor model’s overhead small enough for a high-frequency market-data path? We measured the deployed system in two configurations. In the *one-thread* configuration, the socket-reader thread decodes each datagram and updates the book itself. In the *two-stage* configuration, the reader hands each datagram to a decode-and-book actor on a second thread through an asynchronous mailbox. The one-thread path is the faster of the two and provides the main result; the two-stage path is reported for comparison. Because its mailbox makes queue occupancy observable, the two-stage path also admits a finer decomposition of latency into a floor, an in-packet serialization cost, and a queueing component, which Sections 10.3–10.6 develop.

### 10.1 Setup

The path under measurement is the market-data ingest of Section 8: read a UDP datagram off CME MDP 3.0 multicast, decode the Simple Binary Encoding (SBE) messages [35] inside it, apply them to an order book, and publish. Two timestamps are taken per message —  $t_0$  when the datagram comes up from the socket,  $t_1$  when the book update is published — and their difference is recorded for every message. Because  $t_0$  is a *software* timestamp taken at the socket read, time spent in the NIC, the kernel, and the socket buffer is excluded: the measurement is socket-to-book, not wire-to-book.

On the two-stage path the reader passes each datagram to the decode-and-book actor by an asynchronous `send`. The send is necessarily asynchronous, since a blocking reader would stop draining its socket and drop packets. Decode and book construction then run together on the second thread. The hand-off between the two stages is the decoder’s mailbox, a bounded ring buffer of datagrams (a `BQueue`), and we write `qlen` for its occupancy: the number of datagrams already waiting in the ring when a given datagram is handed to the decoder, so that `qlen = 0` means the ring was empty and the datagram was picked up at once. References below to the ring or the queue are to this mailbox. The hand-off has a single bursty producer, the regime in which `BQueueBatched` is marginally faster than the `BQueue` used here (Section 9.2); the difference is a few percent of the mailbox cost and negligible against the  $\sim 7 \mu\text{s}$  floor of this path. The one-thread path has no hand-off, no ring, and no `qlen`.

Both paths were measured on live CME futures for three instruments — the E-mini S&P 500 (ES), the E-mini Nasdaq-100 (NQ), and the 10-year Treasury Note (ZN) — each with a book-update and a trade stream. Latency on the one-thread path was measured in two five-minute runs, discarding the first minute of each; the receiver used kernel-bypass networking (Onload) on a single feed, with feed B switched off, and threads were not pinned to cores. The two-stage data comes from a single 53-minute session on a busy afternoon, 8.29 million messages in all.

## 10.2 The one-thread path

Table 8 gives the distribution of socket-to-book latency on the one-thread path. The book-update medians are 0.8–1.1  $\mu$ s. This is the cost of decode and book update themselves, the work every message pays when no cross-thread hand-off lies in its path. Against it, a `fast_send` hop ( $\sim 9$  ns, Table 6) is about 1%.

**Table 8.** Socket-to-book latency ( $\mu$ s) on the one-thread path, by stream.

| Stream   | Messages | $p1$ | $p10$ | $p50$ | $p90$ | $p99$ | $p99.9$ |
|----------|----------|------|-------|-------|-------|-------|---------|
| ES book  | 401,462  | 0.5  | 0.7   | 1.1   | 2.7   | 9.9   | 24.1    |
| NQ book  | 581,010  | 0.4  | 0.5   | 0.8   | 1.9   | 4.7   | 8.8     |
| ZN book  | 141,972  | 0.4  | 0.6   | 1.0   | 2.5   | 20.6  | 69.3    |
| ES trade | 45,733   | 0.5  | 0.7   | 2.5   | 8.8   | 36.2  | 70.1    |
| NQ trade | 21,093   | 0.4  | 0.6   | 0.9   | 3.3   | 17.8  | 31.5    |
| ZN trade | 13,282   | 0.4  | 0.7   | 6.6   | 41.5  | 90.5  | 124.0   |

**Comparison with the two-stage path.** On the two-stage path the book-update medians are 7.1–7.3  $\mu$ s (Table 12), about 6  $\mu$ s higher. The main structural difference between the two paths is the cross-thread hand-off through the asynchronous mailbox, and the direction is the one the paper’s own measurements predict: waking a sleeping thread costs microseconds (Table 5). We do not, however, attribute the whole  $\sim 6$   $\mu$ s to the hand-off, because the runs also differ in market load, in length, and possibly in network stack and configuration. The floor and tail reported for the two-stage path below are therefore properties of that path, not the cost of decode and book update.

## 10.3 The per-message model: floor plus serialization

Latency on both paths follows the same per-message model; what differs is the floor. The decode-and-book work itself sets a floor of about 1  $\mu$ s, as the one-thread path shows (Section 10.2). On the two-stage path the floor is higher,  $\sim 7$   $\mu$ s, because each datagram also crosses a thread through the mailbox, and because of the other differences between the runs noted above. The coefficients and the serialization evidence below come from the two-stage run, for which each message’s in-packet position and queue occupancy were recorded.

One UDP datagram carries many SBE messages, decoded in order. Write  $\text{idx}$  for a message’s position within its packet, counted from zero:  $\text{idx} = 0$  is the message decoded first,  $\text{idx} = 1$  the next, and so on. A message at position  $\text{idx}$  therefore waits for the  $\text{idx}$  decodes ahead of it before its own runs — a wait that sits inside its measured latency, and that is zero exactly when  $\text{idx} = 0$ . Restricting to messages that arrive to an empty ring (nothing queued ahead of their packet), so the only remaining wait is in-packet, the median latency is linear in  $\text{idx}$ ,

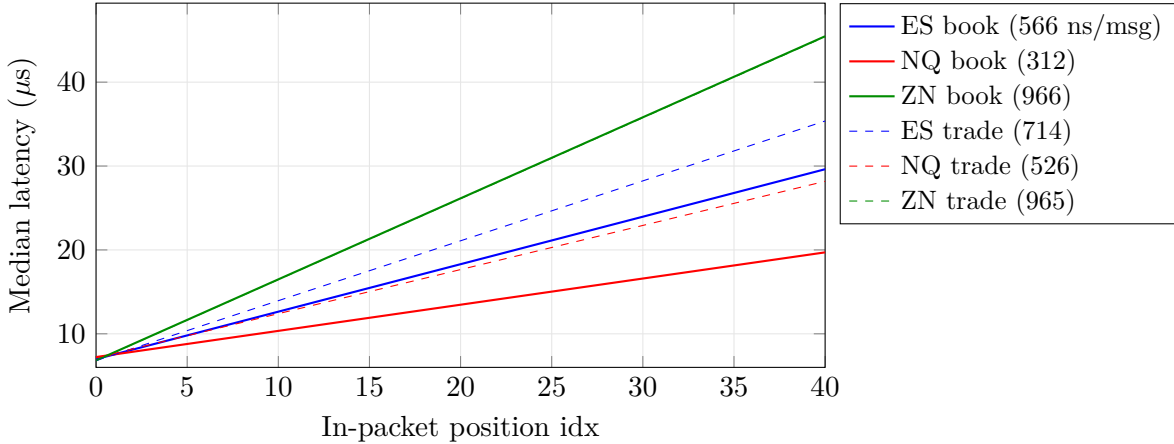
$$\text{median latency} \approx \text{floor} + \text{slope} \times \text{idx},$$

with the two coefficients of Table 9, fitted on medians. (The mean is unsuitable here: a single multi-millisecond stall in a cell moves the mean by tens of microseconds, so every quantity in this section is a median or a percentile.)

The *floor* — the intercept, a message first in its packet — agrees to within 0.42  $\mu$ s across six independent streams, 6.8–7.2  $\mu$ s. It comprises the  $\sim 1$   $\mu$ s of decode-and-book work together with the cross-thread hand-off and whatever else differs between the runs; it is not the cost of decode and book alone. It is insensitive to queue length, and it is stable under load, moving by at most +0.4  $\mu$ s

**Table 9.** The per-message model median  $\approx$  floor + slope  $\times$  idx on the two-stage path, fitted on medians at queue length zero. The floor agrees within  $0.42 \mu\text{s}$  across six streams; the slope, the per-message decode-and-book cost, varies  $\sim 3\times$  and drives the within-packet tail.

| Stream   | Floor ( $\mu\text{s}$ ) | Slope (ns/msg) |
|----------|-------------------------|----------------|
| ES book  | 6.98                    | 566            |
| NQ book  | 7.23                    | 312            |
| ZN book  | 6.83                    | 966            |
| ES trade | 6.81                    | 714            |
| NQ trade | 7.14                    | 526            |
| ZN trade | 6.81                    | 965            |



**Figure 2.** The per-message model median = floor + slope  $\times$  idx (Table 9), one line per stream. All six share a  $\sim 7 \mu\text{s}$  floor at idx = 0; the slope — the per-message serialization cost — sets the divergence, from 312 ns/msg (NQ book) to 966 ns/msg (ZN book).

(and *falling* on two of three instruments) during a Federal Reserve release that raised the packet rate  $4.6\text{--}9.8\times$  in a second. The synchronous per-hop cost is small against either floor: a `fast_send` hop adds  $\sim 7.5$  ns over a direct call ( $\sim 9$  ns in all, about one mispredicted virtual call, Table 6; the full ping/pong round trip is  $\sim 30$  ns), about 1% of the one-thread median. The asynchronous cross-thread hand-off is not small: it contributes to the floor of the two-stage path, and its queue to that path’s tail (Section 10.6).

The *slope* — the per-message service cost within the decode-and-book stage,  $0.31\text{--}0.97 \mu\text{s}$ ; whether it is dominated by SBE decode, by book mutation, or by cache effects we do not resolve here — is the dominant measured contributor to the within-packet component of the tail. In-packet position scales linearly into latency: the last message of a 45-message packet incurs  $44 \times 0.57 \approx 25 \mu\text{s}$  on top of the floor, deterministically. On the trade streams a message is essentially never alone (99.3–99.6% have another message ahead of them in the same packet), which is why the trade medians already sit above the book medians. Shaving 100 ns off the floor moves every message by 100 ns; shaving it off the slope moves a message by  $\text{idx} \times 100$  ns, and idx is largest in the tail. On the two-stage path the design lever for the tail is therefore the slope rather than the mailbox queue, which is rarely occupied: between 87% and 99.7% of messages arrive to find the ring empty. The floor itself is lowered by removing the hand-off, as the  $0.8\text{--}1.1 \mu\text{s}$  book-update medians of the one-thread path show (Table 8).

## 10.4 Three estimates of the floor

The floor of Table 9 is a fitted intercept. Two further estimates, constructed differently, agree with it. The first isolates the hot path directly: the subset of messages with  $q_{len} = 0$  and  $idx = 0$ , for which neither the queue nor in-packet serialization contributes, so its median is the floor with no model interposed (Table 10). Across six streams the spread is  $1.41 \mu s$ ; across the three book streams it is  $0.35 \mu s$ .

**Table 10.** Direct estimate of the floor: median latency ( $\mu s$ ) and sample size of the messages at  $q_{len} = 0$  and  $idx = 0$ , for which neither queueing nor in-packet serialization contributes.

| Stream   | Median ( $\mu s$ ) | $n$       |
|----------|--------------------|-----------|
| ES book  | 7.01               | 1,781,767 |
| NQ book  | 7.24               | 1,841,447 |
| ZN book  | 6.89               | 671,271   |
| ES trade | 7.62               | 1,008     |
| NQ trade | 8.30               | 298       |
| ZN trade | 7.96               | 474       |

The second estimate is the fitted  $idx = 0$  intercept of the median ladder (Table 9). The third is measured during the Federal Open Market Committee statement, when the packet rate rose  $6.5\times$ ,  $4.6\times$ , and  $9.8\times$  on the three instruments within a second; taking the floor from single-message packets (span  $\approx 1$ ) in that window gives the third row of Table 11. The three agree to within a few tenths of a microsecond.

**Table 11.** Three independent estimates of the floor ( $\mu s$ ) for the book streams. The three methods differ in their sample, their estimator, and the load regime they were taken under, and agree to within  $0.4 \mu s$ .

| Method                                            | ES   | NQ   | ZN   |
|---------------------------------------------------|------|------|------|
| Direct median ( $q_{len} = 0, idx = 0$ )          | 7.01 | 7.24 | 6.89 |
| Fitted $idx = 0$ intercept (median ladder)        | 6.98 | 7.23 | 6.83 |
| Floor at span $\approx 1$ during the FOMC release | 7.1  | 7.5  | 7.0  |

The third estimate is the most informative. During the release the floor moved by at most  $+0.4 \mu s$ , and fell on two of the three instruments. The hot path is not rate-sensitive: three methods, three instruments, all within  $6.8\text{--}7.5 \mu s$ . The floor is the most stable quantity in the two-stage measurements — it moved negligibly when the sample tripled, when the estimator changed from mean to median, and when the packet rate rose tenfold.

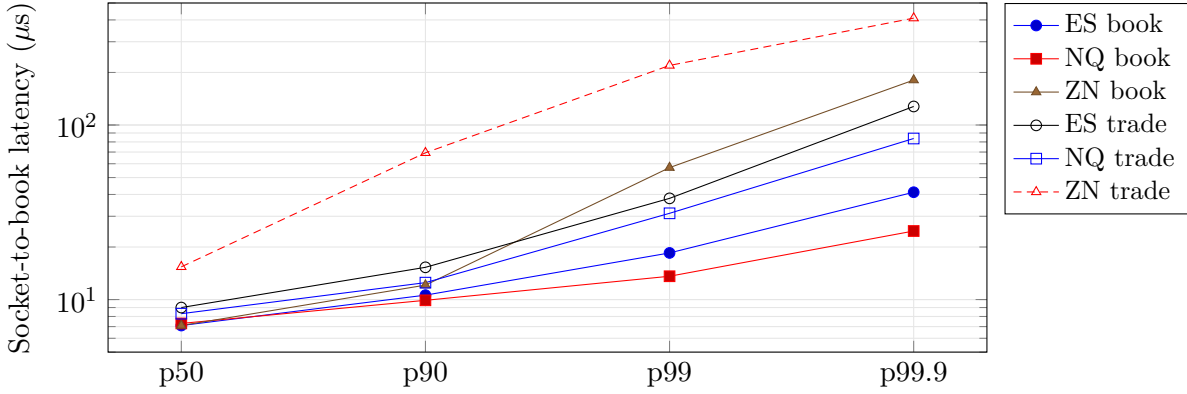
## 10.5 The distribution and its tail

For a trading system the relevant quantity is the tail, not the median. Table 12 gives the unconditional distribution on the two-stage path. The medians cluster — three separate multicast channels, three instruments, all within  $0.2 \mu s$  — but the tails diverge: ES book’s  $p_{99}$  is  $18.5 \mu s$ , ZN book’s is  $57.0 \mu s$ , and ZN trade’s is  $219 \mu s$ . Expressed as amplification from median to  $p_{99}$ , the streams range from  $1.9\times$  (NQ book) to  $14.2\times$  (ZN trade). A single “ $8 \mu s$ ” figure would be accurate for the median but uninformative about the tail. The one-thread path shows the same divergence one level down (Table 8): book-update medians of  $0.8\text{--}1.1 \mu s$ , but a ZN trade  $p_{99}$  of  $90.5 \mu s$ .

Two mechanisms produce this tail, and the next two subsections take them in turn: queue occupancy, when a burst backs up the decoder’s ring (Section 10.6), and in-packet position, when a

**Table 12.** Unconditional socket-to-book latency distribution ( $\mu\text{s}$ ) on the two-stage path, over every admitted message of each stream. The median is uniform across streams; the tail is not.

| Stream   | Messages  | $p50$ | $p90$ | $p99$ | $p99.9$ | max    |
|----------|-----------|-------|-------|-------|---------|--------|
| ES book  | 2,861,519 | 7.1   | 10.6  | 18.5  | 41.2    | 1148.4 |
| NQ book  | 3,792,033 | 7.3   | 9.9   | 13.6  | 24.7    | 5567.0 |
| ZN book  | 1,119,746 | 7.1   | 12.1  | 57.0  | 180.9   | 2458.2 |
| ES trade | 296,736   | 9.0   | 15.3  | 38.0  | 127.7   | 421.5  |
| NQ trade | 110,003   | 8.3   | 12.5  | 31.2  | 83.7    | 694.2  |
| ZN trade | 114,154   | 15.4  | 69.3  | 219.4 | 409.5   | 504.8  |



**Figure 3.** Socket-to-book latency percentiles per stream (Table 12), log scale. Medians cluster near 7–15  $\mu\text{s}$ ; the tails diverge, ZN trade reaching 219  $\mu\text{s}$  at  $p99$  and 410  $\mu\text{s}$  at  $p99.9$ .

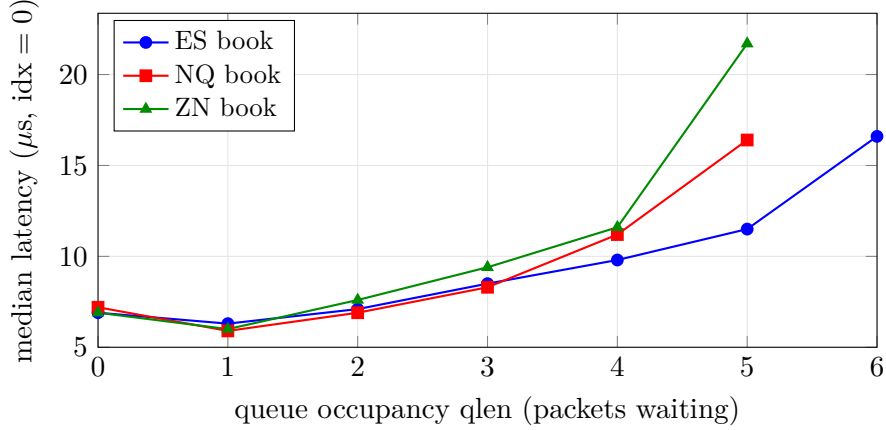
datagram carries many messages (Section 10.7). The first exists only because the two-stage path hands datagrams across threads through a mailbox. Their full deconfounding, and how to mitigate the tail, is in a companion paper [39].

## 10.6 Latency versus queue occupancy

Recall that  $qlen$  is the number of datagrams already waiting in the decoder’s ring when a datagram is handed to it, so that  $qlen = 0$  means an empty ring. Figure 4 plots the median socket-to-book latency at each queue length for the three book streams, with in-packet position held at  $idx = 0$  so the curve isolates the queue from packet size. Two features stand out.

*At  $qlen = 0$  the path is at its floor.* The median sits at the  $\sim 7 \mu\text{s}$  baseline of Section 10.3, and this is the state 87% to 99.7% of messages find: an empty ring adds no queueing wait. For the overwhelming majority of messages the path runs at that baseline.

*As the ring fills, latency rises, and convexly.* Each additional queued packet costs more than the last: ES book climbs from 6.9  $\mu\text{s}$  at  $qlen = 0$  to 16.6  $\mu\text{s}$  at  $qlen = 6$ , and ZN book from 6.9 to 21.7  $\mu\text{s}$  at  $qlen = 5$ . This is ordinary queueing — when a burst arrives faster than the decoder drains it, packets wait behind packets, and the wait grows with occupancy — and it is why the latency tail of the two-stage path is, in part, a queue-occupancy effect of the cross-thread hand-off. It is a *rare* effect:  $qlen \geq 3$  occurs on only 0.08% of ES book messages (2,311 of 2,861,519), so the queue is a large effect at the far percentiles on a small fraction of messages, not a small effect everywhere. Its full deconfounding from packet size is carried out in the companion paper [39].



**Figure 4. Latency rises with queue occupancy.** Median socket-to-book latency against queue occupancy qlen for the three book streams, with in-packet position held at  $\text{idx} = 0$  to isolate the queue from packet size. At  $\text{qlen} = 0$  — the state 87%–99.7% of messages find — the median sits at the  $\sim 7 \mu\text{s}$  floor; each further queued packet adds more than the last (convex), so the tail, where it is queue-driven, tracks qlen. The effect is confined to a rare event ( $\text{qlen} \geq 3$  on 0.08% of messages). Deep-queue points ( $\text{qlen} \geq 5$ ) rest on few messages and are noisier.

## 10.7 Suitability: the per-hop cost and the hand-off

The framework enters the measured latency in two different ways, and they differ by orders of magnitude. The synchronous per-hop cost is small: one `fast_send` dispatch costs about one mispredicted virtual call ( $\sim 9 \text{ ns}$ , Table 6), about 1% of the  $\sim 1 \mu\text{s}$  book-update median of the one-thread path (Section 10.2) and about 0.1% of the  $\sim 7 \mu\text{s}$  floor of the two-stage path. It does not grow with in-packet position. The asynchronous cross-thread hand-off is not small. It is the main structural difference between the  $\sim 7 \mu\text{s}$  medians of the two-stage path and the 0.8–1.1  $\mu\text{s}$  medians of the one-thread path (Section 10.2), so it contributes to the floor, and through its ring it contributes to the tail (Section 10.6).

The tail of the two-stage path has two sources. First, *queue occupancy*: when a burst backs the ring up, latency rises with qlen (Section 10.6) — a genuine tail contributor, but a rare one, since 87% to 99.7% of messages find an empty ring; it exists because of the cross-thread hand-off. Second, and for most messages the larger effect, *in-packet position*: a datagram carries several SBE messages decoded in sequence, so a message late in a large packet waits behind the earlier ones. In-packet position is a property of the feed — bursts of market events that the exchange coalesces into large packets — and applies on either path. The full characterization of the arrival process, and what it implies for mitigating the tail, is the subject of a companion paper [39].

The verdict is affirmative and specific, and it is the paper’s case for `fast_send` and groups: the per-hop cost of synchronous delivery is about 1% of the decode-and-book work, while an asynchronous cross-thread hand-off in the path raises the floor and adds a queueing tail. Keeping the path on one thread brings the book-update median to about 1  $\mu\text{s}$ . What an implementer must engineer is where the path crosses threads, the per-message service time — the decode-and-book slope — and the feed’s batch structure.

## 10.8 Limits of these measurements

Because  $t_0$  is a software timestamp at the socket read, both sets of measurements are socket-to-book rather than wire-to-book; hardware receive timestamps would close that gap and were not taken.

The two-stage data comes from one machine and one 53-minute session, and the one-thread data from two short runs in a separate session, on a single feed with threads not pinned. Because the two paths were not measured side by side, the  $\sim 6 \mu\text{s}$  between their medians is not attributed to the hand-off alone. On the two-stage path the per-message service cost varies  $\sim 3\times$  across instruments (ZN is dearest), and whether that is cold-cache residency or a genuinely heavier message mix cannot be separated without hardware counters. And a separate population of multi-millisecond stalls — the largest latencies in the two-stage data (the `max` column of Table 12) — is real and remains unexplained here.

## 11 Production and simulation: a duality of the group

**The group defines a single message order.** A `Group` co-schedules its member actors on one thread behind one shared mailbox (Section 9.1): the members do not own separate queues; the group’s single queue holds every message addressed to any member, and the run loop dispatches them one at a time. Beyond the latency effect measured earlier — the collapse of the round trip from  $\sim 3.4 \mu\text{s}$  to 90 ns — this has a structural consequence. A single queue imposes a single total order on all messages entering the group, so the group’s entire execution — every actor state transition and every outgoing message — is a deterministic function of that one input sequence.

**The same actors run in production and in simulation.** That property makes one group two systems. In production the queue is fed by live sources — socket readers, timers, market-data actors — and the run loop executes in real time. In simulation the queue is fed from a recorded or synthetic sequence, and the run loop executes as fast as the host permits. The member actors are identical in both modes: the framework gives a handler nothing that distinguishes a message from a live socket from one read from a backtest file, which is the receiver transparency of Section 7.2 extended from the delivery mechanism to the data source. As there, this is a property of the interface, not isolation the language enforces. No strategy actor is edited, recompiled, or conditionally branched to move between live trading and backtest. Given the same ordered input sequence and deterministic injected services, replay reproduces the same state transitions and outputs — the single shared queue fixes the message order, and the actors add no other source of nondeterminism. That is precisely the one discipline the duality imposes: an actor must acquire time, randomness, and external input through messages or injected interfaces rather than through global calls, so that the simulator can supply them deterministically.

**Relation to prior work.** Two ingredients of this design have prior art; their combination is uncommon. The single-shared-queue group is a first-class facility in `Actor4j` [11], where a group’s actors run on one thread with their mailboxes outsourced to the thread’s queue — presented there as a message-passing throughput optimization, not as a simulation mechanism. The same-code production/simulation duality is the principle of deterministic simulation testing: `FoundationDB`’s `Flow` runtime [37] and systems such as `TigerBeetle` run unmodified production code inside a single-threaded deterministic simulator, but there the unit of determinism is an asynchronous continuation or an explicit state machine rather than a group of mailbox actors sharing a queue. Agent-based market simulators such as `ABIDES` [38] drive actor-like agents behind one time-ordered event queue, but those agents exist only in simulation and are not a deployed production system. The design here couples the two: the shared-queue group is what makes mode-agnostic actors run identically in real-time production and in deterministic backtest. We are not aware of this specific combination being described elsewhere.

**Example: developing an execution algorithm.** This duality is how the order-level shadowing execution algorithm of the companion paper [36] was developed. The strategy is an actor that consumes market-data and order-event messages and emits child-order messages; it was built and validated entirely in simulation, driven from recorded CME sessions, and then deployed to production without modification — the same actor and the same handlers, reading from a live feed instead of a file. The 246-session slippage comparison reported there was produced by the simulation mode of the same group that runs the strategy live. The child-order logic is written once and is agnostic to the mode in which it executes.

## 12 Discussion and limitations

**The scope is co-located actors.** `fast_send` applies only when sender and receiver share an address space; a synchronous cross-machine call would reintroduce the network round trip it exists to avoid, and the actor model’s location transparency means the sender can fall back to asynchronous delivery when the receiver is remote. Receiver transparency is what makes that fallback free: the same handler serves the remote asynchronous case and the local synchronous case.

**The cost of transparency.** Heap-allocating the reply in both modes is a deliberate cost paid to keep the handler unable to distinguish them. The optimized synchronous-only reply removes that allocation but forfeits transparency and must therefore be confined to code paths statically known to be synchronous; mixing it with asynchronous delivery is an error the mechanism reports rather than tolerates.

**Blocking runs counter to conventional guidance.** Established actor frameworks discourage blocking inside actors precisely because it can starve thread pools and cause deadlock [7, 6, 10]. `fast_send` blocks the sender by construction, and its safety rests on the cyclic-invocation detector and on the availability of the non-blocking and best-effort variants for paths that cannot tolerate an unbounded wait. Where a caller must never block, `fast_send_x` and `fast_send_void` provide bounded-time and never-blocking alternatives behind the same handler interface. Even so, the call-chain detector addresses only cycles on one thread (Section 7.3), and only *cyclic* waits; avoiding thread-pool starvation for acyclic blocking patterns — a chain that blocks on an actor whose handler is itself waiting on a scarce resource — remains a burden on the developer that asynchronous delivery does not impose.

**Contention on a hot actor.** A synchronous send holds the receiver’s exclusive-access lock for the duration of handler execution, so concurrent synchronous sends to the same heavily contended actor serialize on that lock, and the blocking variants stall their caller threads while they wait. Under heavy contention this can perform worse than asynchronous delivery, in which senders enqueue and continue and the messages buffer without stalling the callers. The non-blocking (`fast_send_x`) and best-effort (`fast_send_void`) variants bound or remove the stall — the latter by falling back to the mailbox — so the ratio of synchronous completions to asynchronous fallbacks doubles as a signal that a target has become hot enough that the asynchronous path is preferable.

**Permissive detection shifts a burden to the programmer.** The (actor, message-class) refinement admits callbacks by allowing an actor’s state to be re-entered by a nested handler, which reintroduces the reentrancy hazards that strict per-actor sequential processing removes. A nested handler can observe and mutate the actor’s state mid-update, which relaxes the actor model’s

guarantee of single-threaded sequential state mutation and admits the invariant violations that guarantee is designed to prevent. The conservative policy avoids this at the cost of rejecting some legitimate callback structures; the choice between them is exposed as configuration rather than fixed.

**Further evaluation.** The microbenchmarks of Section 9.1 establish the per-operation latency reductions and the two-order-of-magnitude tail benefit of the memory pool directly. The natural next step measures the mechanism under load: synchronous versus asynchronous *throughput* across contention levels, chain depths, and message sizes on the x86-64 Linux target, together with the distribution of synchronous completions versus asynchronous fallbacks for `fast_send_void`. That saturated, multi-axis characterization is the subject of ongoing work.

## 13 Future work

1. **Detecting cycles across threads** (Section 7.3), which a thread-local call chain cannot see: a global wait-for graph, a global lock order, or a bounded try-lock with asynchronous fallback.
2. **The saturated-throughput and contention evaluation** (Section 12).
3. **Completing kaspar-hft:** the cyclic-invocation detector and its refinement, the three remaining delivery operations, the optimized synchronous-only reply, and enforcement of a single reply per message.

## 14 Conclusion

The actor model has long been dismissed for high-frequency trading on a plausible but mistaken premise: that its safety is inseparable from a discipline — one thread per actor, a mailbox per actor, a message and a context switch for every interaction — whose per-message cost cannot fit a microsecond budget. The premise conflates the model with one way of implementing it. For *co-located* actors the discipline can be honored without that cost, and the resulting system is fast enough for the HFT hot path while keeping the properties that motivate the model: per-actor state isolation, freedom from data races, and sequential single-message reasoning. The synchronous path gives up part of the asynchronous model’s freedom from deadlock: a cycle of synchronous calls spread across threads deadlocks. This paper supports that thesis both analytically and with a deployed, measured implementation — *kaspar-hft* — rather than as theory alone.

Asynchronous message passing is intrinsic to the actor model’s guarantees and indispensable across machines, but for co-located actors it imposes heap allocation, queue operations, scheduling, and context switches that isolation does not require. `fast_send` provides a synchronous alternative in which the sender’s thread runs the receiver’s handler directly under the receiver’s exclusive-access lock and takes the reply as a value, while the handler’s code does not depend on the delivery mode, the executing thread, or the sender’s blocked state — receiver transparency. A thread-local call-chain membership test, checked before lock acquisition, catches synchronous cycles on one thread before they deadlock or overflow the stack; cycles across threads need a different mechanism. A refinement keyed on (actor, message-class) pairs admits legitimate callbacks while still catching unbounded recursion, and five delivery operations expose the latency/robustness spectrum behind one transparent handler interface. The performance case is made analytically, as the elimination of per-message work, and corroborated by microbenchmarks in which the synchronous path completes

a round trip in tens of nanoseconds and a memory pool cuts the latency tail by two orders of magnitude.

The applied study places these numbers against a live market-data path. On the kaspar-hft framework, with the socket reader decoding each packet and updating the book on its own thread, socket-to-book book-update medians are 0.8–1.1  $\mu$ s, and the framework’s per-hop cost ( $\sim 9$  ns, about one unpredictable virtual call) is about 1% of that. When the reader instead hands each datagram to a second thread through an asynchronous mailbox, latency decomposes into a  $\sim 7$   $\mu$ s floor and a per-message serialization slope, the floor stable to within a few tenths of a microsecond across three estimators and a tenfold change in packet rate. The hand-off is the main structural difference between the two paths, though the runs also differ in load and configuration, so we do not attribute the whole difference to it. On the two-stage path part of the tail is mailbox-queue backlog; the dominant effect is the market’s arrival process, which is non-Poisson and strongly clustered, and converts bursts into large packets and in-packet position into latency through the slope — characterized in a companion paper [39]. The parameters to optimize are therefore where the path crosses threads and the per-message service time, not the message rate. The same study also fixes the choice among the four mailbox queue implementations — BQueue by default, ShardedBQueue for high-fan-in actors — and shows that a group of actors sharing one queue is at once a real-time execution engine and a deterministic simulator, so a trading strategy is developed in backtest and deployed to production without modification.

Taken together, the mechanism and the measurement answer the question in the title for the workload we tested. The four extensions — `fast_send`, actor groups, per-actor mailbox selection, and the memory pool — reduce the per-hop cost of the actor framework on a live tick-to-book path to about one percent of the  $\sim 1$   $\mu$ s decode-and-book work when that path stays on one thread; an asynchronous cross-thread hand-off in the path, by contrast, raises the floor and adds a queueing tail. For this co-located market-data workload, kept on one thread, the actor discipline therefore carries a small per-hop latency cost, and in return the design retains data-race freedom, single-message reasoning (which we argue, but do not here demonstrate, is amenable to automatic code generation), and a production/simulation duality in which the same strategy code is backtested and traded without change. These results suggest that actor-based designs need not be excluded from HFT systems on the grounds of per-message messaging overhead — long the stated reason — and that a deployed system runs on one. What this study does not establish, and what remains for future work, is behavior under saturation and contention: the evaluation here is sequential, single-machine, and socket-timestamped, not a throughput or multi-core stress test. The remaining work is listed in Section 13.

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